

Connected Weather Station

Raspberry Pico 2 WH

Hardware Assembly & Wiring Guide

Read each step in order — every step builds on the previous one

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Project	L3 SPI — Connected Weather Station
Platform	MicroPython on Raspberry Pico 2 WH
Document	Hardware Assembly & Wiring Guide

Bill of Materials

Component	Qty	Notes
Raspberry Pi Pico 2 WH	1	Male headers soldered
Breadboard 830-tie	1	Power rails on both sides
BME280 module (I2C)	1	JST-SH STEMMA QT port
VEML7700 module (I2C)	1	Two JST-SH STEMMA QT ports
OLED SSD1306 128×32	1	I2C, address 0x3C
Analogue cup anemometer	1	Voltage output, magnetic sensor
LM335 sensor (TO-92)	1	3 legs — only 2 used
Tactile push button (4-pin)	1	Small breadboard type
 LED blue	1	5 mm standard
 LED yellow	1	5 mm standard
 LED white	1	5 mm standard
 LED red	1	5 mm standard
Resistors 220 Ω	4	One per LED
Resistor 2.2 kΩ	1	LM335 bias pull-up
JST-SH STEMMA to male headers	1	Pico W → VEML7700
JST-SH STEMMA QT 4-pin cable	1	VEML7700 → BME280
Dupont male-female cables	4	Pico W → OLED (I2C)
Jumper wires	~15	Breadboard interconnects

Assembly Steps

Step 1 — Mount the Pico 2 WH on the Breadboard

Insert the Pico 2 WH straddling the centre of the breadboard so that:

- **Pin 1** (GP0) is at the top-left.

- **Pin 40** (VBUS) is at the top-right.
- Both rows of pins overhang into the central tie-point area.

Warning: Orient the Pico with the USB connector facing the top of the breadboard. Do not force it — each leg must fall cleanly into its own hole.

Step 2 — Ground Rails (GND) — Black Wires

The breadboard has two blue (–) power rails. Connect both from two separate GND pins on the Pico WH.

Source (Pico WH)	Destination	Colour
Pin 3 — GND	Left blue rail (–)	Black
Pin 38 — GND	Right blue rail (–)	Black

Warning: These two wires are the foundation of the entire build. Without them, nothing will work.

Verification: Both blue (–) rails are now tied to the Pico W ground.

Step 3 — Power Rails — Red Wires

Distribute two voltage levels across the red (+) power rails.

Source	Destination	Voltage	Colour
Pin 40 — VBUS	Red rail (+) upper zone	5 V (USB)	Red
Pin 36 — 3V3(OUT)	Red rail (+) other side	3.3 V	Orange

Warning: VBUS (5 V) is only present when the Pico is powered via USB. The 3.3 V rail is regulated by the Pico at all times.

Warning: All sensors operate at 3.3 V. The 5 V rail is used only as a bias supply for the LM335 temperature sensor.

Verification: Left red rail = 3.3 V. Right red rail = 5 V.

Step 4 — LM335 Temperature Sensor + 2.2 kΩ Resistor

The LM335 (TO-92, 3 legs) is biased with a simple pull-up circuit. Only the (+) anode and (–) cathode legs are used. The centre ADJ leg is left floating.

Circuit schematic:

```
5V -> [2.2k ohm] -> node A -> LM335 anode (+) -> LM335 cathode (-) -> GND
|
GP27 (ADC1, Pin 32)
```

Breadboard connections:

- **2.2 kΩ resistor:** one leg into the 5 V rail, the other into a free hole (node A).
- **LM335 anode (+):** into node A (same row as resistor output).
- **LM335 cathode (–):** into a hole connected to the GND rail.
- **GP27 wire:** from node A to **Pin 32** (GP27).

Warning: Flat face of TO-92 on the left (legs toward you): cathode (–) = leftmost leg, anode (+) = rightmost leg.

Warning: The centre ADJ leg must be left floating. Do not connect it.

Warning: GP27 is rated 3.3 V maximum. The bias circuit keeps node A safely below this ceiling at all normal ambient temperatures.

Verification: 5 V → 2.2 kΩ → node A (shared with GP27 and anode) → cathode → GND.

Step 5 — Cup Anemometer (Wind Speed) → GP26

The anemometer outputs an analogue voltage that varies with rotational speed.

Anemometer wire	Connection
VCC (+)	3.3 V red rail
GND (–)	Blue GND rail
Signal OUT	Pin 31 — GP26 (ADC0)

Verification: Anemometer signal on GP26 (ADC0), not GP27 (ADC1 — reserved for LM335).

Step 6 — Status LEDs + 220 Ω Resistors

Four LEDs indicate the system status at a glance. Each is wired identically: GPIO → 220 Ω → anode (+) → cathode (–) → GND.

LED	GPIO	Pin	Blinks when...
 Blue	GP13	Pin 17	Wi-Fi is connected (once per 12 s cycle)
 Yellow	GP14	Pin 19	Node-RED is active (once per 12 s cycle)
 White	GP10	Pin 14	Adafruit IO publish succeeded
 Red	GP15	Pin 20	Sensor out of range or MQTT failure

Important rules:

-  Blue blinks whenever Wi-Fi is up, even when Node-RED is the active channel.
-  White only blinks when Adafruit IO is the active channel. When Node-RED is connected, Adafruit IO is skipped and white stays off.
-  Red blinks on sensor errors and MQTT failures. Wi-Fi being absent is **not** an error and does not trigger the red LED.
- All blinks last **0.5 seconds**.

Warning: Long leg = anode (+) on the GPIO/resistor side. Short leg = cathode (–) on the GND side.

Verification: GPIO → 220 Ω → anode → cathode → GND. Same circuit × 4.

Step 7 — Push Button → GP6

The button is wired directly between GP6 and GND. The firmware enables the internal pull-up — no external resistor is required.

Button leg	Connection
Leg A (or 1)	Pin 9 — GP6
Leg B (or 2)	Blue GND rail

Three press durations are recognised:

Duration	Threshold	Action on OLED
Short	< 1 s	5-slide sensor carousel (2 s per slide, animated progress bar)
Long	1 s to 5 s	System status: SD / Wi-Fi / cloud state (4 s)
Extra	≥ 5 s	Error detail: each sensor error shown for 2 s, or “All systems OK”

Warning: A 4-pin tactile button is symmetric: legs 1 & 2 are joined, legs 3 & 4 are joined. Use one leg from each side of the symmetry axis.

Warning: At rest the pin reads HIGH due to the pull-up. Pressing drives it to LOW.

Verification: One leg on GP6 (Pin 9), the other on GND. No external resistor needed.

Step 8 — JST STEMMA QT Cable: Pico 2 WH → VEML7700

Use the JST-SH STEMMA to male-headers cable. The JST end plugs into the VEML7700; the male headers go into the breadboard.

Wire colour	Signal	Pico 2 WH	Pin
Black	GND	GND rail	—
Red	3.3 V	3V3 rail	—
Blue	SDA	GP4	Pin 6
Yellow	SCL	GP5	Pin 7

Warning: The JST-SH connector is keyed — it can only be inserted one way. Do not force it.

Verification: VEML7700 powered and on the I2C bus (address 0x10).

Step 9 — JST STEMMA QT Cable: VEML7700 → BME280

The VEML7700 has two JST-SH ports. Port 1 is used by Step 8. Connect port 2 to the BME280 (I2C daisy-chain).

Pico W (master) -> VEML7700 (0x10) -> BME280 (0x76)

Warning: The BME280 has only one JST port. The OLED is connected separately in Step 10.

Verification: Full I2C chain: Pico W → VEML7700 → BME280 complete.

Step 10 — OLED SSD1306 128×32 — Dupont Wires

The OLED has no JST port. Connect it with 4 male-to-female Dupont wires, in parallel on the same I2C bus as the two sensors.

OLED pin	Signal	Destination	Pin
VCC (or VDD)	3.3 V	3.3 V rail	Pin 36
GND	GND	Blue GND rail	—
SDA	SDA	GP4 SDA row	Pin 6
SCL	SCL	GP5 SCL row	Pin 7

Warning: GP4 and GP5 now each carry three connections (STEMMA + Dupont + Dupont). This is correct — I2C is a shared bus.

Warning: The OLED is powered off at boot to save energy. It wakes only on button press and powers off again automatically.

Verification: OLED address 0x3C. VCC = 3.3 V, GND, SDA = GP4, SCL = GP5.

Step 11 — Micro-SD Card Module — SPI Bus 0

The SD card module communicates over SPI bus 0.

SD pin	Signal	GPIO	Pin
VCC	3.3 V	3.3 V rail	—
GND	GND	GND rail	—
MISO	SPI MISO	GP16	Pin 21
CS	SPI CS	GP17	Pin 22
SCK	SPI CLK	GP18	Pin 24
MOSI	SPI MOSI	GP19	Pin 25

Warning: Pin order varies between SD modules. Always verify against your module's silkscreen.

Warning: Format the micro-SD as FAT32 before first use. The firmware creates `meteo.csv` automatically on first boot.

Verification: SPI bus 0, baudrate 4 MHz, polarity = 0, phase = 0, MSB first.

Complete Pin Reference

Phys. pin	GPIO	Type	Connected to
Pin 6	GP4	I2C SDA	I2C bus: VEML7700 + BME280 + OLED (parallel)
Pin 7	GP5	I2C SCL	I2C bus: VEML7700 + BME280 + OLED (parallel)
Pin 9	GP6	GPIO IN	Push button (internal pull-up) → GND
Pin 14	GP10	GPIO OUT	220 Ω →  white LED → GND
Pin 17	GP13	GPIO OUT	220 Ω →  blue LED → GND
Pin 19	GP14	GPIO OUT	220 Ω →  yellow LED → GND
Pin 20	GP15	GPIO OUT	220 Ω →  red LED → GND
Pin 21	GP16	SPI MISO	SD card module — MISO
Pin 22	GP17	SPI CS	SD card module — CS
Pin 24	GP18	SPI SCK	SD card module — SCK
Pin 25	GP19	SPI MOSI	SD card module — MOSI
Pin 31	GP26	ADC0	Anemometer — analogue signal output
Pin 32	GP27	ADC1	LM335 — node A (2.2 kΩ to 5 V / read point)
Pin 36	3V3	3.3 V	3.3 V power rail + all I2C modules
Pin 3	GND	Ground	Left blue (—) rail
Pin 38	GND	Ground	Right blue (—) rail
Pin 40	VBUS	5 V	5 V rail (LM335 bias resistor)

LED Behaviour Reference

LED	GPIO	Blinks when...
 Blue	GP13	Wi-Fi is connected at the time of the 12-second sending cycle. Blinks regardless of which output channel is active.
 Yellow	GP14	Node-RED heartbeat is detected over USB. When active, Adafruit IO is automatically skipped.
 White	GP10	Adafruit IO MQTT publish succeeded. Does not blink when Node-RED is active.
 Red	GP15	A sensor reading is outside its valid range, a sensor fails to respond, the Node-RED heartbeat is unexpectedly lost, or an MQTT connection / publish failed. Wi-Fi being absent is not an error.

Sensor	Valid range
Temperature	−20.0 °C to +60.0 °C
Humidity	0.0 % to 100.0 % RH
Pressure	870.0 hPa to 1085.0 hPa
Luminosity	0 lux to 120 000 lux
Wind speed	0.0 km/h to 200.0 km/h

Pre-Power Checklist

Verify every item before connecting the USB cable.

- ☐ Both GND rails fed from Pin 3 and Pin 38 of the Pico W
- ☐ 3.3 V (Pin 36) and VBUS (Pin 40) each on their own red rail
- ☐ LM335: 2.2 k Ω from 5 V to node A; GP27 on node A; cathode on GND
- ☐ Anemometer signal on GP26 (Pin 31), not GP27
- ☐ 4 LEDs each with a 220 Ω resistor on the GPIO side; cathode (short leg) on GND
- ☐  Blue on GP13 (Pin 17),  yellow on GP14 (Pin 19),  white on GP10 (Pin 14),  red on GP15 (Pin 20)
- ☐ Button: one leg on GP6 (Pin 9), the other on GND
- ☐ STEMMA cable from Pico on port 1 of VEML7700; BME280 on port 2
- ☐ OLED: VCC = 3.3 V, GND, SDA = GP4, SCL = GP5
- ☐ SD module: MISO = GP16, CS = GP17, SCK = GP18, MOSI = GP19
- ☐ secrets.py filled in with Wi-Fi SSID, password, and Adafruit IO key

All boxes checked? Connect the USB cable and watch the LEDs.
